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HONORS CHEMISTRY • UNIT 4

Chemical Reactions

Write, balance, classify and predict the products of chemical reactions, then show only the particles that actually change.

HOW TO USE THIS PACKET

Write in pencil: you will balance a lot of equations. Use the blank space for particle sketches. LAB CORNER boxes are things you would see at a lab bench. Extra practice and a review are at the back.

What we will learn

4.1	Balancing equations & writing reactions
4.2	Types of reactions
4.3	Predicting products: synthesis, decomposition, combustion
4.4	Predicting products: single replacement (activity series)
4.5	Predicting products: double replacement (solubility rules)
4.6	Net ionic equations
4.7	Naming & writing formulas (review)

By the end of this unit I can...

- Balance equations using coefficients and explain why equations must be balanced.
- Translate between word equations and balanced symbol equations with state symbols.
- Classify a reaction as synthesis, decomposition, single replacement, double replacement or combustion.
- Predict the products of synthesis, decomposition and combustion reactions.
- Use the activity series to predict single replacement reactions and solubility rules to predict precipitates.
- Write complete, ionic and net ionic equations and identify spectator ions.
- Name and write formulas for ionic compounds, molecular compounds, acids and polyatomic ions.

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4.1 Balancing Chemical Equations & Writing Reactions

Law of conservation of mass • coefficients • state symbols

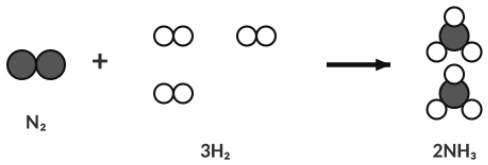
BIG IDEA

Atoms are not created or destroyed in a chemical reaction, so every atom on the **reactant** side must appear on the **product** side. We balance by changing **coefficients** (the big numbers in front), never the subscripts.

Key vocabulary — write each definition in your own words

Term	Definition (your words)	Example / sketch
Reactant / product	*Which side of the arrow?*	
Coefficient vs subscript	*Which one can you change when balancing?*	
Law of conservation of mass	*What does it say about atoms?*	
State symbols	*What do (s), (l), (g) and (aq) mean?*	

1. Why does a chemical equation need to be balanced? What does a balanced equation represent? (Think about a cooking recipe.)



$N_2 + 3H_2 \rightarrow 2NH_3$: 2 N and 6 H on each side.

HOW TO BALANCE (TRY THIS ORDER)

1. Write the correct formulas (diatomic elements: **H_2, N_2, O_2, F_2, Cl_2, Br_2, I_2**).
2. Make an atom tally for each side. Treat a polyatomic ion (PO_4, NO_3, SO_4 …) as one unit if it stays together.
3. Balance metals first, then non-metals, then **H, then O last**. Elements that appear in one compound on each side are easiest.
4. If you get a fraction, multiply everything by 2. Re-check every atom. Coefficients must be the lowest whole numbers.

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WORKED EXAMPLE — A COMBUSTION

Balance: ___ C_2H_6 + ___ O_2 → ___ CO_2 + ___ H_2O

Work it out here:

Practice: balance each equation

Use the space under each equation for your atom tally (count each element on both sides) and your trial coefficients.

	Balance each equation. Show your atom tally and work below it, then write the final balanced equation.
a.	___ Na + ___ O_2 → ___ Na_2O
b.	___ K + ___ B_2O_3 → ___ K_2O + ___ B
c.	___ H_3PO_4 + ___ KOH → ___ K_3PO_4 + ___ H_2O
d.	___ Al + ___ S_8 → ___ Al_2S_3

	Balance each equation. Show your atom tally and work below it, then write the final balanced equation.
e.	___ N_2 + ___ O_2 → ___ N_2O_5
f.	___ H_3PO_4 + ___ $Mg(OH)_2$ → ___ $Mg_3(PO_4)_2$ + ___ H_2O
g.	___ NH_3 + ___ O_2 → ___ N_2 + ___ H_2O
h.	___ FeS_2 + ___ O_2 → ___ Fe_2O_3 + ___ SO_2

2. Draw particle diagrams to balance $H_2 + O_2 \rightarrow H_2O$. Draw the correct number of molecules on each side.

Particle sketch:

Writing equations from words

HELPFUL REMINDERS

- Diatomic elements are always written as molecules: H_2, N_2, O_2, F_2, Cl_2, Br_2, I_2.
- Use the ion charges to write correct formulas (criss-cross) and then balance.
- Add states: (s), (l), (g), (aq).

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Polyatomic ion	Name	Polyatomic ion	Name
NH_4^+		SO_4^{2-}	
NO_3^-		CO_3^{2-}	
OH^-		PO_4^{3-}	
$C_2H_3O_2^-$		ClO_3^-	
MnO_4^-		CN^-	

Reaction described in words	Balanced equation with states
Sodium metal reacts with chlorine gas to produce sodium chloride.	
Aluminum metal reacts with aqueous copper(II) nitrate to produce aqueous aluminum nitrate and copper metal.	
Aqueous nitric acid reacts with aqueous magnesium hydroxide to produce aqueous magnesium nitrate and liquid water.	
Propane gas (C_3H_8) reacts with oxygen gas to produce gaseous carbon dioxide and gaseous water.	
Solid aluminum oxide decomposes into solid aluminum and oxygen gas.	
Solid ammonium chloride decomposes into gaseous nitrogen trihydride and gaseous hydrochloric acid.	
Aqueous lead(II) nitrate and aqueous potassium iodide react to produce solid lead(II) iodide and aqueous potassium nitrate.	

Turn each equation into a word equation:

Equation	Word equation
$Na_2O(s) + H_2O(l) \rightarrow 2NaOH(aq)$	
$CS_2(l) + 3O_2(g) \rightarrow CO_2(g) + 2SO_2(g)$	
$K(s) + CuNO_3(aq) \rightarrow KNO_3(aq) + Cu(s)$	
$2Li(s) + 2H_2O(l) \rightarrow 2LiOH(aq) + H_2(g)$	

4.2 Types of Reactions *Six patterns*

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Type	General pattern	Clue	Example
Synthesis			
Decomposition			
Single replacement			
Double replacement			
Neutralization			
Combustion			

Acids and bases: how do I tell them apart?

Key terms — write each definition in your own words

Term	Definition (your words)	Example / sketch
Acid	*What does it release in water? What does its formula start with?*	
Base	*What does it release in water? What does its formula usually end with?*	
Salt	*What is left after the acid and base react?*	

> **NEUTRALIZATION**
> An acid and a base swap partners: the H^+ from the acid joins the OH^- from the base to make **water**, and the leftover ions form a **salt**. **acid + base → salt + water**

Predict the salt and water

Reactants	Balanced equation	Name of the salt
HCl + NaOH		
H_2SO_4 + KOH		
HNO_3 + $Ca(OH)_2$		
HBr + $Al(OH)_3$		

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Classify each reaction

#	Reaction	Type
1	$C_5H_{12} + 8O_2 \rightarrow 5CO_2 + 6H_2O$	
2	$2K_3PO_4 + 3BaCl_2 \rightarrow Ba_3(PO_4)_2 + 6KCl$	
3	$2Al(OH)_3 + 3H_2SO_4 \rightarrow Al_2(SO_4)_3 + 6H_2O$	
4	$2AgNO_3 + Na_2CO_3 \rightarrow Ag_2CO_3 + 2NaNO_3$	
5	$2Al_2O_3 \rightarrow 4Al + 3O_2$	
6	$2Al + 6HCl \rightarrow 3H_2 + 2AlCl_3$	
7	$C_4H_{10}O + 6O_2 \rightarrow 4CO_2 + 5H_2O$	
8	$3HBr + Al(OH)_3 \rightarrow 3H_2O + AlBr_3$	
9	$2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$	
10	$2HgO \rightarrow 2Hg + O_2$	
11	$Na_2O + H_2O \rightarrow 2NaOH$	
12	$Cu + 2AgNO_3 \rightarrow 2Ag + Cu(NO_3)_2$	

1. Which reaction type has the pattern AB → A + B?

A) synthesis B) decomposition C) single replacement D) combustion

2. $HNO_3 + KOH \rightarrow KNO_3 + H_2O$ is…

A) synthesis B) decomposition C) neutralization D) combustion

4.3 Predicting Products: Synthesis, Decomposition & Combustion *Reactants → products*

PREDICTION RULES

Synthesis (two elements combine): a **metal + non-metal** makes an ionic compound. Use the group number to find each charge (Group 1 → 1+, Group 2 → 2+, Group 3 → 3+; Group 15 → 3−, Group 16 → 2−, Group 17 → 1−), then criss-cross the charges to write the formula ($Mg^{2+} + O^{2-} \rightarrow MgO$). (If a problem combines two non-metals, the product's formula will be given.)

Decomposition: a binary compound (two elements) breaks apart into its two **pure elements** (by electrolysis or strong heat). Remember the diatomic elements: H_2, N_2, O_2, F_2, Cl_2, Br_2, I_2.

Combustion: hydrocarbon (or an alcohol) + $O_2 \rightarrow CO_2 + H_2O$ (complete combustion).

Complete and balance

	Reactants →	Balanced equation (products and coefficients)	Type
1	$Na + Cl_2 \rightarrow$		

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	Reactants →	Balanced equation (products and coefficients)	Type
2	$Mg + O_2 \rightarrow$		
3	$Al + S_8 \rightarrow$		
4	$Li + N_2 \rightarrow$		
5	$NaCl \rightarrow$		
6	$H_2O \rightarrow$		
7	$Al_2O_3 \rightarrow$		
8	$CH_4 + O_2 \rightarrow$		
9	$C_2H_6 + O_2 \rightarrow$		
10	$C_2H_4 + O_2 \rightarrow$		

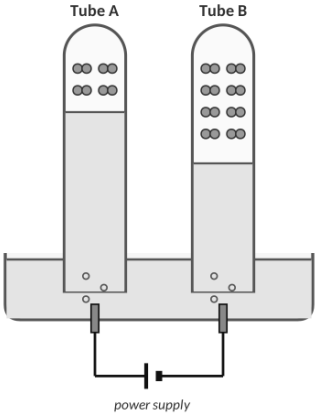
1. Propane (C_3H_8) burns completely in oxygen. Which products form?

A) CO and H_2 B) CO_2 and H_2O C) C and H_2O D) C_3H_8 and O_2

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LAB CORNER • Electrolysis of water

A power supply is connected to two electrodes in water. Gas collects in two inverted tubes in a volume ratio of 2 : 1.



Each pair of dots is one gas molecule. Both tubes were filled by the same power supply for the same time.

2. Look at the amounts of gas collected. (a) Which tube holds hydrogen and which holds oxygen? Explain how you decided. (b) Write the balanced decomposition equation for water.

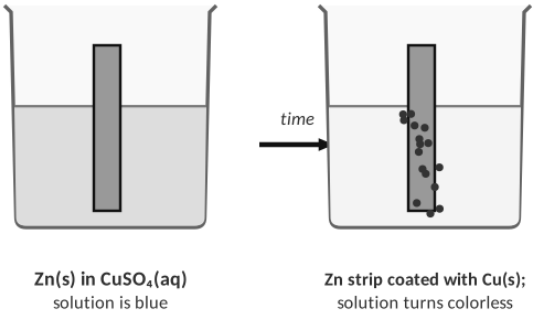
Answer:

4.4 Single Replacement & the Activity Series

A more active element replaces a less active one

BIG IDEA

In a single replacement reaction a free element swaps places with an element in a compound. It only works if the free element is **more active** (more eager to lose or gain electrons) than the one it replaces.



Zinc is more active than copper: $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$.

Activity series	Most active → least active
Metals	Li, K, Ba, Sr, Ca, Na, Mg, Al, Zn, Fe, Ni, Pb, (H), Cu, Ag, Au
Halogens	$F_2 > Cl_2 > Br_2 > I_2$
Reacts with cold water?	Li, K, Ba, Sr, Ca, Na → metal hydroxide + H_2 ($2Na + 2H_2O \rightarrow 2NaOH + H_2$); metals below Ca, Na do not.

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1. Explain in words how to decide whether a single replacement reaction will happen.

2. Write the balanced equation for $Zn + CuSO_4$ and tell which metal is more active and how you know.

Predict the products (write NR if no reaction)

	Reactants	Products (balanced) or NR
1	$Li + BaCl_2 \rightarrow$	
2	$Li + CaS \rightarrow$	
3	$K + BaCl_2 \rightarrow$	
4	$Ba + NaF \rightarrow$	
5	$Na + MgCl_2 \rightarrow$	
6	$F_2 + NaCl \rightarrow$	
7	$Cl_2 + NaF \rightarrow$	
8	$Cu + FeSO_4 \rightarrow$	
9	$Na + H_2O \rightarrow$	
10	$Ag + H_2O \rightarrow$	

3. Which will NOT react?

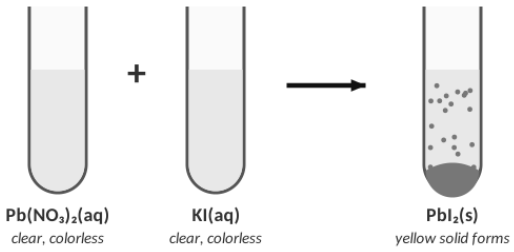
A) $Zn + CuSO_4$ B) $Mg + FeCl_2$ C) $Cu + ZnCl_2$ D) $Cl_2 + KBr$

4.5 Double Replacement & Solubility Rules *Swap partners • precipitate, water or gas*

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BIG IDEA

In a double replacement reaction two compounds exchange partners: AB + CD → AD + CB. In water the reaction only "happens" if one new compound is a **solid (precipitate)**, **water** or a **gas**; otherwise the ions just stay mixed (no reaction).



Two clear solutions form a solid precipitate (colored yellow in real life).

Solubility rules (water)	**Soluble (aq) …**	**… except**
Group 1 ions (Li^+, Na^+, K^+) and NH_4^+		
Nitrates (NO_3^-), acetates ($C_2H_3O_2^-$), chlorates		
Chlorides, bromides, iodides		
Sulfates		
Hydroxides, carbonates, phosphates, sulfides, sulfites		

STEPS

1. Swap the cations. 2. Write each new formula using ion charges (criss-cross). 3. Balance. 4. Use the solubility rules to label (s) or (aq). 5. If both products are aqueous: **no reaction**.

#	**Reactants**	**Products (balanced, with states) or NR**
11	$BaCl_2 + H_3PO_4$ (aq) →	
12	$K_2SO_4 + MgF_2$ (aq) →	
13	$AlCl_3 + BaSO_4$ (aq) →	
14	$Li_2S + AgNO_3$ (aq) →	
15	$BaCl_2 + Na_2CO_3$ (aq) →	

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#	Reactants	Products (balanced, with states) or NR
16	$Al_2(SO_4)_3 + Na_3PO_4$ (aq) →	
17	$Na_2SO_4 + NH_4I$ (aq) →	
18	$Li_3PO_4 + BaCl_2$ (aq) →	
19	$AlBr_3 + H_3PO_4$ (aq) →	
20	$Zn(NO_3)_2 + Ba(OH)_2$ (aq) →	

1. Which pair gives a precipitate?

A) $NaCl + KNO_3$ B) $Pb(NO_3)_2 + KI$ C) $NH_4Cl + NaNO_3$ D) $K_2SO_4 + NaCl$

2. What is the precipitate when $AgNO_3$(aq) and NaCl(aq) are mixed?

A) $NaNO_3$ B) AgCl C) $AgNO_3$ D) no precipitate

4.6 Net Ionic Equations *Show only the particles that change*

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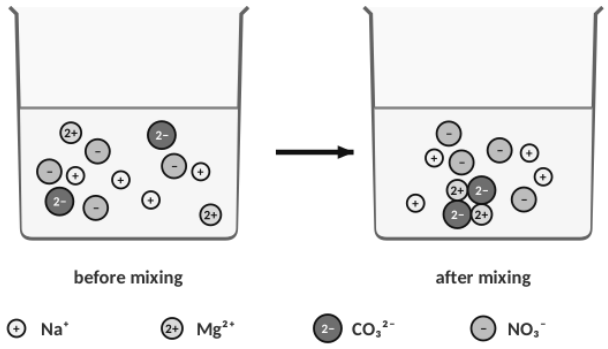
Key terms

Term	Definition (your words)	Example / sketch
Net ionic equation	*What does it represent?*	
Spectator ion	*What do the species that do NOT react get called?*	

HOW TO WRITE ONE

1. Balanced molecular equation with states.
2. Split **soluble ionic compounds (aq)** into ions. Keep solids (s), liquids (l), gases (g) and weak electrolytes together.
3. Cancel the ions that appear unchanged on both sides (spectators).
4. Write what is left; check atoms AND charge.

A student dissolves sodium carbonate and magnesium nitrate in water. The diagrams show the beaker before and after the reaction.



1. (a) Write the balanced equation. (b) Which ions react and which are spectators? (c) Justify with the diagram. (d) Write the net ionic equation.

Work:

Write the net ionic equation

	Reaction	Net ionic equation

a	$SrBr_2(aq) + K_2SO_4(aq) \rightarrow SrSO_4(s) + 2KBr(aq)$	
b	$3Ca(OH)_2(aq) + 2FeCl_3(aq) \rightarrow 3CaCl_2(aq) + 2Fe(OH)_3(s)$	
c	$2AgOH(s) + H_2CO_3(aq) \rightarrow Ag_2CO_3(s) + 2H_2O(l)$	
d	$Fe(s) + NiCl_3(aq) \rightarrow FeCl_3(aq) + Ni(s)$	
e	Sodium sulfide(aq) + iron(II) nitrate(aq) →	

2. Spectator ions are…

A) ions that form the precipitate
B) ions that stay dissolved and unchanged
C) ions that gain electrons
D) ions in the solid

4.7 Naming & Writing Formulas (Review) *Ionic • molecular • acids*

Type	Rule	Example
Ionic (metal + non-metal)		
With polyatomic ion		
Molecular (2 non-metals)		
Binary acid		
Oxyacid		

	Given	Name or formula
1	Li_2S	
2	beryllium phosphide	
3	$Be(OH)_2$	
4	P_2Br_4	
5	SF_6	
6	magnesium acetate	
7	diboron tetrahydride	

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	Given	Name or formula
8	dinitrogen trioxide	
9	hydrocyanic acid	
10	lithium iodide	
11	$KMnO_4$	
12	$HClO_3$	

1. Which is the correct name for N_2O_3?

A) nitrogen oxide B) dinitrogen trioxide C) nitrogen(III) oxide D) dinitrogen oxide

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PRACTICE

Extra Practice — Unit 4

Show your work.

A. Balance

	Balance each equation (show your atom tally below it)
a	$Al + HCl \rightarrow AlCl_3 + H_2$
b	$Fe + O_2 \rightarrow Fe_2O_3$
c	$C_4H_{10} + O_2 \rightarrow CO_2 + H_2O$
d	$Ca(OH)_2 + H_3PO_4 \rightarrow Ca_3(PO_4)_2 + H_2O$

B. Classify and predict

Reactants	Balanced products	Type
$Mg + O_2$		
HgO (heat)		
$C_3H_8 + O_2$		
Zn + HCl		

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Reactants	Balanced products	Type
$AgNO_3 + CaCl_2$		
$HCl + Ca(OH)_2$		

C. Precipitation lab data

A student mixes pairs of 0.1 M solutions and records whether a solid forms. Predict the result and write the net ionic equation for the ones that react.

Solutions mixed	Precipitate?	Net ionic equation
$NaCl + AgNO_3$		
$KI + Pb(NO_3)_2$		
$NaOH + CuSO_4$		
$KNO_3 + NaCl$		
$BaCl_2 + K_2SO_4$		

D. Think

1. A student balances $H_2 + O_2 \rightarrow H_2O$ by changing it to $H_2 + O_2 \rightarrow H_2O_2$. Explain what is wrong with this method.

2. Explain why no reaction occurs when NaCl(aq) and KNO_3(aq) are mixed.

3. Which is a correct net ionic equation for $AgNO_3 + NaCl$?

A) $AgNO_3 \rightarrow Ag^+ + NO_3^-$ B) $Ag^+ + Cl^- \rightarrow AgCl(s)$ C) $Na^+ + NO_3^- \rightarrow NaNO_3$ D) $Ag + Cl \rightarrow AgCl$

4. Which reaction type is: $2Mg + O_2 \rightarrow 2MgO$?